Robert Garcia

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Education

Harvard University — Cambridge, MA
Bachelor of Science: Mechanical Engineering. GPA: 3.55 — May 2027
Relevant Coursework: Computer Aided Machine Design, Thermodynamics, Engineering Design Seminar, Solid Mechanics, Mechanical Systems, Materials, Data Structures and Algorithms.
Design Project: Aerobic Charge: Converting kinetic energy during exercise to electrical charge.

Hartford High School — Hartford, CT
Salutatorian. Honors Science Program. 1st place winner, Robot Challenge. — June 2023

Engineering Research Experience

Harvard University Laboratory for Nanoscale Optics — Cambridge, MA
Research Assistant — June – Aug 2025

- Optimize the dry etch recipe for titanium dioxide and polycrystalline diamond thin films.
- Fabricate photonic devices (ridges, ring resonators, microdisks, photonic crystals) in TiO2 and polycrystalline diamond films.
- Characterize the fabricated photonic devices using a transmission setup.

National Nanotech Research Center — Pasadena, CA
Research Experience for Undergraduates Intern — June – Aug 2024
Project: Microfabrication of Thin-film Heaters to Simulate Hotspots.

- Fabricated devices for testing the effectiveness of a nanoscale patch used to cool down hotspots on microprocessors.
- Utilized cleanroom facilities to create devices that contain a hotspot heater and temperature sensors to simulate heat generation.

Additional Experience

Harvard School of Engineering and Applied Sciences — Cambridge, MA
Course Assistant — Sept 2025 – present

- Hold weekly office hours, providing tutoring and problem set instruction for students in Computer Science 50: Introduction to Computer Science.

Leadership and Activities

Harvard College Engineering Society, Vice President — Sept 2024 – present

- Organize monthly meetings and activities for the engineering community at Harvard.
- Host guest speakers, manage communications, and plan events.

Projects and Skills

Projects: Electric Bicycle: designed electrical and mechanical systems for custom 1kW bike
Smart Appliance: IoT toaster based on Raspberry Pi, Python code, & 3D-printed body

Laboratory: Cleanroom fabrication experience, including photolithography, e-beam lithography, metallization, thin film deposition, etching, metrology, SEM imaging.

Technology: Advanced HTML, CSS and programming in C, PHP, JavaScript, Python.
Intermediate MATLAB, SolidWorks, Adobe Creative Suite, Google SketchUp.

Languages: Spanish (intermediate), German (beginner).